

FTM IMCQ 6

ANSWER KEY

Pharmacology, Biochemistry, Genetics & Histology

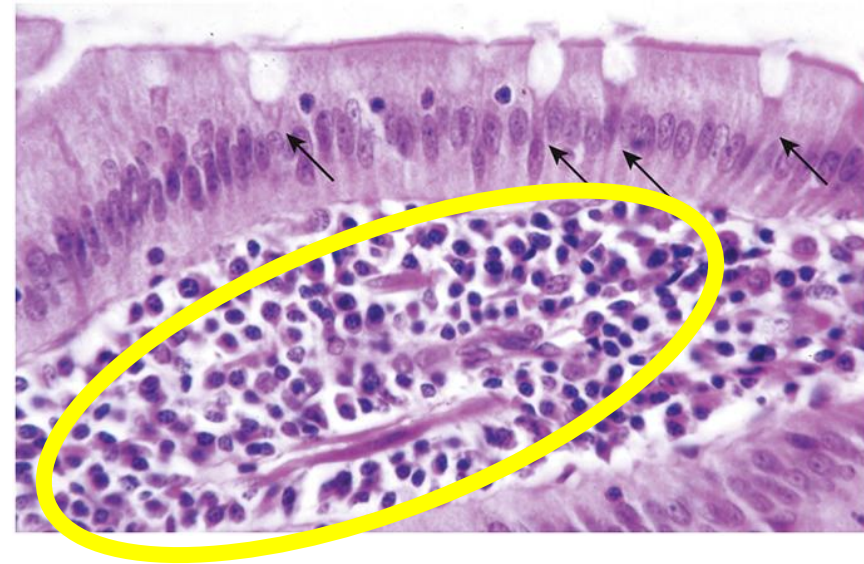
BPM1 FTM

Fall Term 2025

HCB (3 Questions)

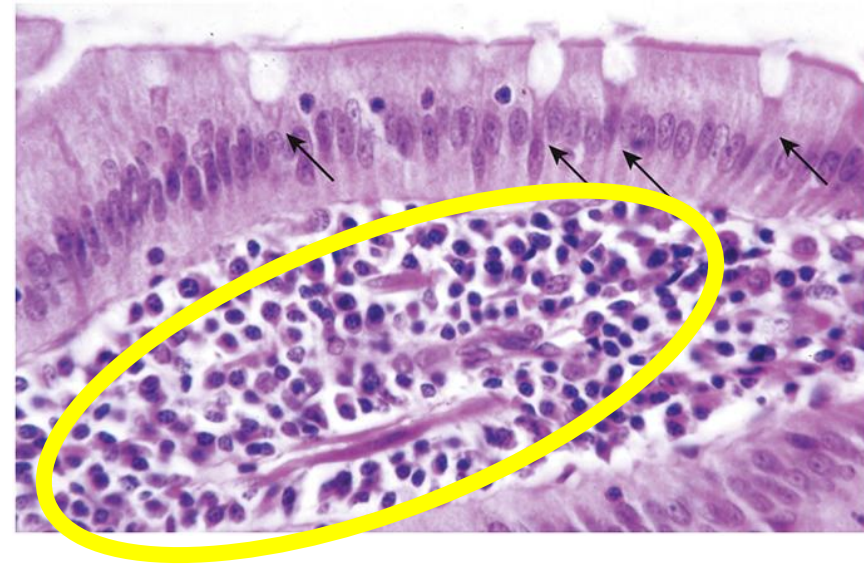
1. A 65-year-old man is brought to the emergency department because of a 1-month history of worsening abdominal pains, constipation, and unintentional weight loss of 6.8-kg (15-lb). Rectal examination shows a mass in the distal rectum that bleeds easily. Tissue samples of the mass obtained for biopsy shows a breach in the basement membrane with infiltration into the tissue outlined in the attached photomicrograph. Which of the following is most likely a function of the tissue into which the mass has infiltrated?

- A. Allow uni-directional stress
- B. Support to the overlying epithelium
- C. Permit stress in multiple directions
- D. Give rise to all other connective tissue types
- E. Insulation and energy storage



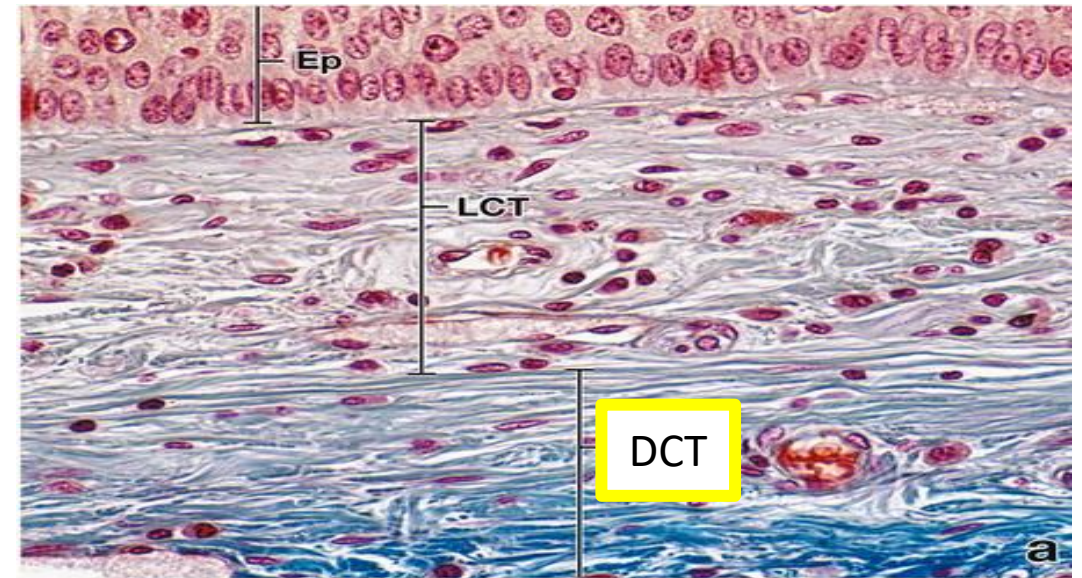
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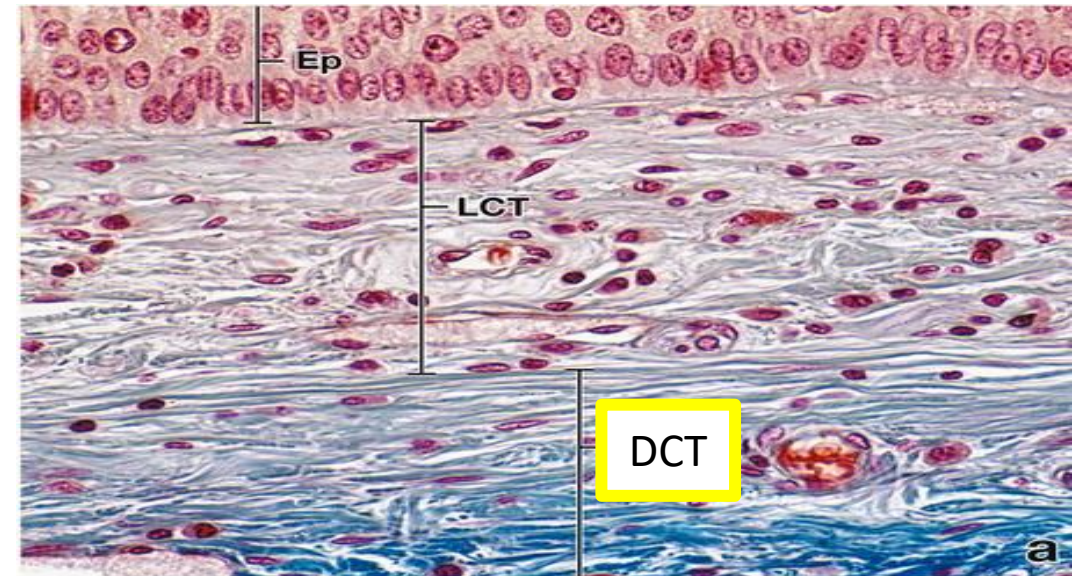
2. A 51-year-old alcoholic man is brought to the emergency department because of a 1-hour history of severe vomiting of bright red blood. Investigations lead to the diagnosis of esophageal varices secondary to portal hypertension. These varices are often found within the walls of the esophagus in regions like the area labeled “DCT” in the attached photomicrograph. Which of the following best characterizes this tissue type?

- A. Also found lining the mucous membrane of the urinary tract
- B. Gel-like ECM composed mainly of hyaluronan
- C. Abundant fibers and fewer cells oriented in one direction
- D. Randomly arranged abundant fibers and fewer cells
- E. Abundant ground substance with spindle-shaped cells



2. A 51-year-old alcoholic man is brought to the emergency department because of a 1-hour history of severe vomiting of bright red blood. Investigations lead to the diagnosis of esophageal varices secondary to portal hypertension. These varices are often found within the walls of the esophagus in regions like the area labeled “DCT” in the attached photomicrograph. Which of the following best characterizes this tissue type?

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3. A 20-year-old man comes to the physician because of a 1-week history of a painful skin rash. He received a bee sting to the same site 2 weeks ago. Physical examination shows mild swelling and redness of the affected area. Examination of biopsy specimens obtained from the rash are studied using different stains. One of the stains used was toluidine blue. When the toluidine blue was applied to the specimen, the granules within the cytoplasm of some cells in the connective tissue was observed to express different colors. Which of the following connective tissue cells are most likely responsible for this reaction?

- A. Fibroblasts
- B. Macrophages
- C. Mast cells
- D. Myofibroblasts
- E. Adipocytes

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Distinguish between the resident cell types found in connective tissue (fibroblasts, myofibroblasts, macrophages, adipocytes, stem cells and mast cells) based on their microanatomical features and functions.SOM.MK.I.BPM1.1.FTM.3.HCB.0856

Biochemistry (4 Questions)

A 13-year-old boy is brought to the physician because of a 2-day history of vision changes. He keeps bumping into other students while walking to class. Physical examination shows a boy with a long face; long, thin fingers; and a prominent sternum. Visual acuity shows 20/50 OD (right eye) and 20/20 OS (left eye). Pupils are equal and reactive bilaterally. Extraocular muscles are intact. There is vibration of the right iris during left and right movement indicating lens dislocation. A diastolic murmur is heard in the aortic area. Which of the following is the most likely function of the deficient protein in this patient?

- A. Glycosylation of hydroxylysine residues on pro-alpha chains
- B. Hydroxylation of proline and lysine residues on pro-alpha chains
- C. Covalent cross linking of tropocollagen
- D. Elastin scaffold formation
- E. Oxidatively deaminating lysine residues of tropoelastin

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An 18-year-old woman is brought to the physician for a follow-up examination. She had three leg bone fractures when she was 3 to 5 years old. She has a family history of a genetic disorder. She is 151 cm (4 ft 11 in) tall, with short arms. Physical examination shows blue sclerae. Targeted genetic testing shows a pathogenic mutation in her *COL1A2* gene resulting in Gly769Arg in the COL1A2 protein. Which of the following best describes the effect of this mutation?

- A. Disruption of crosslinking
- B. Disruption of hydrogen bonding
- C. Distortion of the triple helix
- D. Haploinsufficiency
- E. Loss of gene expression
- F. Disruption of desmosine cross links

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A 76-year-old woman is brought to the physician for a follow-up wound dressing. She sustained a right arm laceration, which was sutured one week ago. She lives alone and is having difficulty caring for herself. She has a limited diet that consists mainly of cookies, crackers, and tea. Physical examination shows poor wound healing. Skin examination shows multiple bruises. Her gingiva is receded and swollen. Which of the following steps of collagen synthesis is most likely impaired in this patient?

- A. Glycosylation of hydroxylated lysine residues
- B. Assembly of three pro- α chains into a triple helix
- C. Hydrogen bond formation in triple helix
- D. Cross-linking of assembled collagen fibrils
- E. N-terminal propeptide cleavage

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A 45-year-old woman comes to the physician because of a 2-week history of pain in both knee joints and her back. She is a farm worker and has been working 6 days a week, lifting loads up to 40 lb (18.1 kg) at various times during the day for the past 10 years. Physical examination shows swollen knee joints that are tender to palpation, with crepitus on both passive and active motion. She is prescribed a supplement containing a glycosaminoglycan, which is non-sulfated and not covalently attached to a protein. Which of the following best describes this supplement?

- A. Keratan sulfate
- B. Hyaluronic acid
- C. Chondroitin sulfate
- D. Dermatan sulfate
- E. Heparan sulfate

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Genetics (4 Questions)

A 14-year-old girl is brought to the physician by her father because of a 3-month history of impaired balance and involuntary jerking and writhing movements. There is limited family history as her father was adopted, and her mother died in a motor vehicle collision 3 years ago. Early-onset Huntington's disease is suspected. If the incidence of Huntington's disease is 1/10,000, which of the following is most likely her risk of being homozygous for the mutant allele?

- A. 1/10,000
- B. 1/5,000
- C. 1/4
- D. 1/400,000,000
- E. 1/25,000,000



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A 32-year-old man and his partner come to the physician for preconception counseling. They are concerned about Gaucher Disease, an autosomal recessive disorder, because of their family histories. The man's oldest nephew is affected. The woman's maternal aunt is affected. They are unrelated and otherwise healthy. Which of the following best describes their risk of having an affected child?

A. 1 (100%)

B. $\frac{1}{2}$

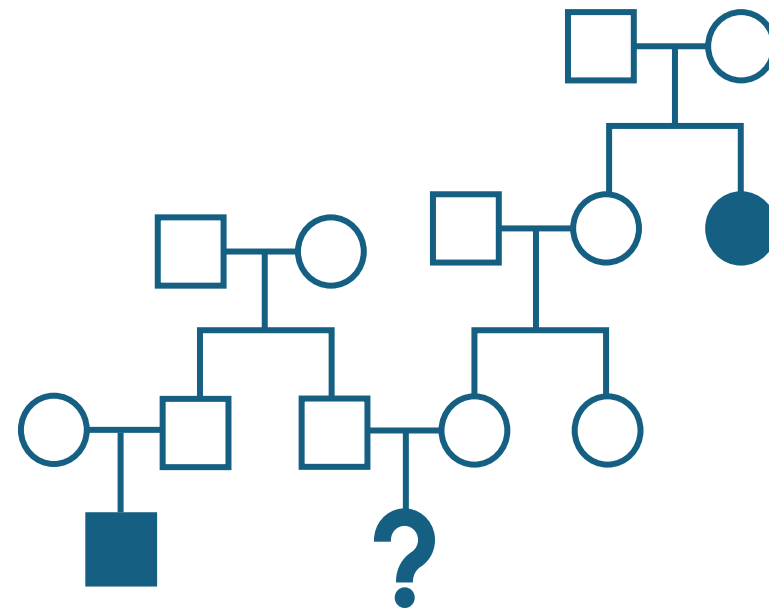
C. $\frac{1}{3}$

D. $\frac{2}{3}$

E. $\frac{1}{4}$

F. $\frac{1}{16}$

G. $\frac{1}{24}$



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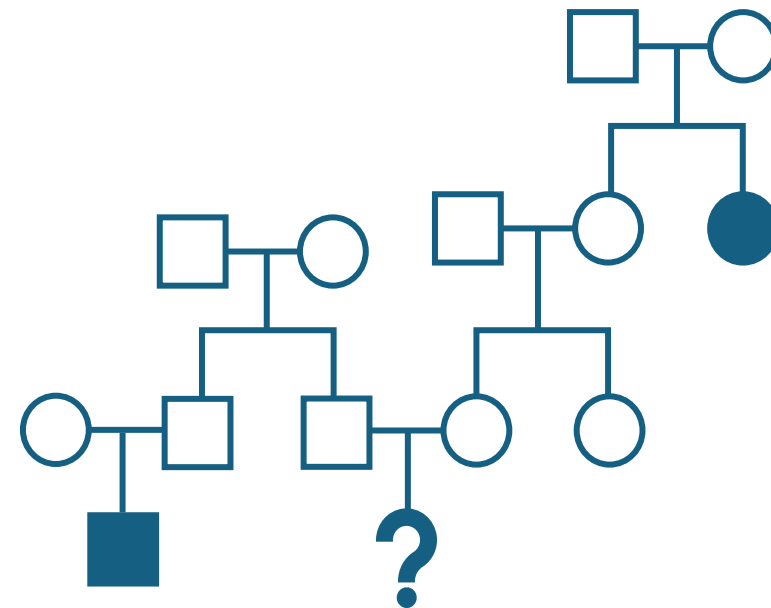
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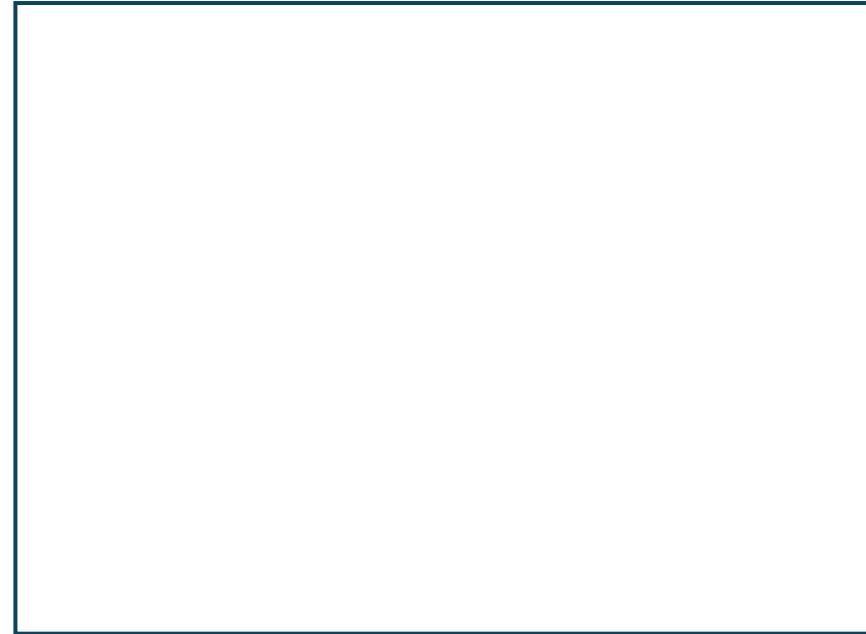
✓ G. $\frac{1}{24}$



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Estimate a carrier frequency from the
Hardy Weinberg equation and
population incidence for one individual
to estimate recurrence risk when that
person mates with an individual who
has familial carrier risk.

A 28-year-old man and his wife come to the physician for genetic counseling. He has a mutation in the factor VIII gene, resulting in low levels of clotting Factor VIII, and has repeated episodes of hemarthrosis. His wife is 24 weeks pregnant with their first child. Ultrasonography shows she is carrying a male. The couple is concerned about the male's risk of having hereditary Factor VIII deficiency. The incidence of the disorder in the population is 1 in 15,000. Which of the following is most likely the mother's risk of being a carrier of the disorder?

- | | |
|-------------------------------|-----|
| A. $\frac{1}{4}$ | 14% |
| B. $\frac{1}{7,500}$ | 14% |
| C. $\frac{1}{2}$ | 14% |
| D. $\frac{2}{3}$ | 14% |
| E. $\frac{1}{\sqrt{1/15000}}$ | 14% |
| F. $\frac{1}{(15,000)^2}$ | 14% |
| G. $\frac{1}{15,000}$ | 14% |



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- F. $\frac{1}{(15,000)^2}$ 14%
- G. $\frac{1}{15,000}$ 14%

SOM.MKIII.BPM1.1.FTM.3.GNET.0088;
Estimate a carrier frequency from the Hardy Weinberg equation and population incidence for one individual to estimate recurrence risk when that person mates with an individual who has familial carrier risk.

A 32-year-old woman and her partner, 34-year-old male, both of Ashkenazi Jewish descent, come to the physician to discuss their results from an Ashkenazi Jewish (AJ) Carrier Testing Panel. They are both carriers of the most common disease-causing allele in their ancestral population, a four-base-pair duplication in the HEXA gene which is associated with the infantile-lethal form of Tay-Sachs disease. The physician tells the couple, as carriers, they will not develop the disease, but they have a 25% chance of having an affected child, and the high frequency of Tay-Sachs carriers may be due to natural selection. Which of the following evolutionary mechanisms best explains the persistence of this disease-causing allele in this population?

- 20% A. Genetic drift due to population bottlenecks
- 20% B. Founder effect in a small population
- 20% C. Heterozygote advantage due to partial resistance to tuberculosis
- 20% D. Increased reproductive fitness in carriers
- 20% E. Non-random mating

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Pharmacology (5 questions)

The pharmacodynamic properties of a new investigational drug for hypertension are being investigated. Isolated strips of vascular smooth muscle are exposed to norepinephrine. In the absence of the investigational drug, norepinephrine produces a dose-dependent increase in smooth muscle contraction. When increasing concentrations of the investigational drug are added, the maximal contractile response to norepinephrine is markedly reduced, the drug binds to an alternate site on the receptor and the concentration of norepinephrine required to produce 50% of the maximal effect remains unchanged. Which of the following best describes the mechanism of action of this investigational drug?

- A. Competitive antagonism
- B. Non-competitive agonism
- C. Partial agonism
- D. Inverse agonism
- E. Physiologic antagonism

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A 12-year-old boy is brought to the physician by his parents because of a 2-day history of abdominal cramps, watery diarrhea and vomiting. He has associated fever. He does not have any recent travel history, and his vaccinations are up to date. His temperature is 38°C (100.4°F), pulse is 110/min and respirations are 18/min. Physical examination shows a dehydrated patient with dry mucous membranes and decreased skin turgor. Abdominal examination shows hyperactive bowel sounds. A diagnosis of acute gastroenteritis is made, and the boy is prescribed oral ibuprofen. Ibuprofen is a weak organic acid with a pK_a of 4.5. Given that the pH of the duodenum is 5.5, which of the following is the best approximate percentage of a given dose of ibuprofen that will be in the liposoluble (unionized) form in the duodenum?

- A. 0.01%
- B. 1%
- C. 10%
- D. 50%
- E. 99%

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- E. 99%

pH – pK	$\log \frac{[\text{unprotonated}]}{[\text{protonated}]}$	$\frac{[\text{unprotonated}]}{[\text{protonated}]}$	% unprotonated	% protonated
3	3	10^3	99.9	0.1
2	2	10^2	99	1
1	1	10	90	10
0	0	1	50	50
-1	-1	10^{-1}	10	90
-2	-2	10^{-2}	1	99
-3	-3	10^{-3}	0.1	99.9

The pharmacodynamic properties of two beta₁-adrenergic antagonists, Drug A and Drug B are being investigated. The two drugs are administered intravenously IV experimental animals and the effect on heart rate is recorded. It is observed that 80 mg of Drug A evoke the same decrease in heart rate as 10 mg of Drug B. Which of the following best describes the properties of Drug A?

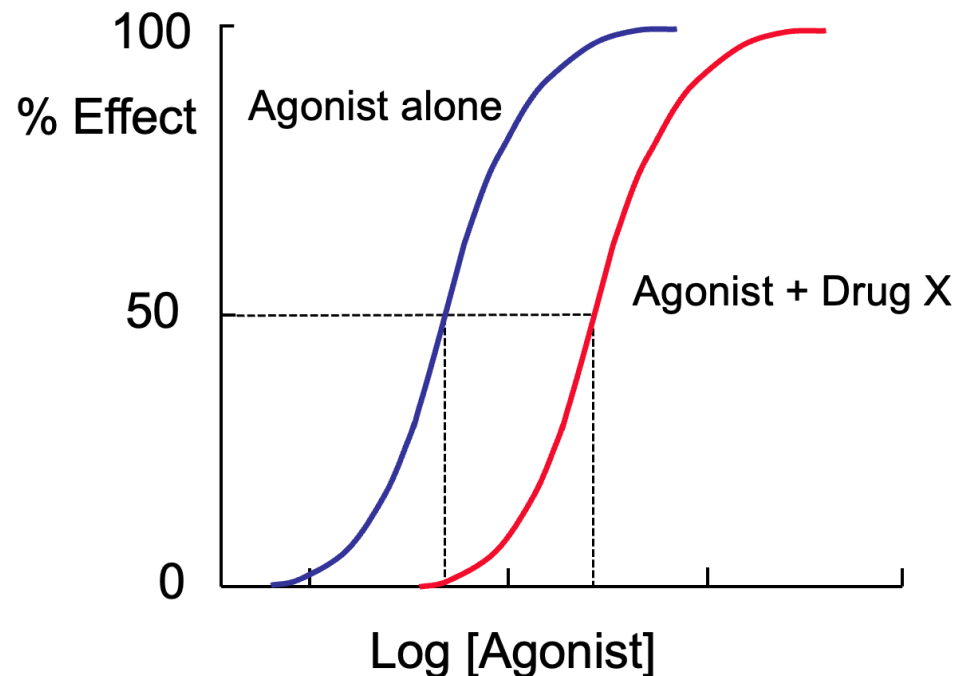
- A. Drug A has a higher therapeutic index than drug B
- B. Drug A has a lower bioavailability than drug B
- C. Drug A has a lower potency than drug B
- D. Drug A has a shorter half-life than drug B
- E. Drug A has a lower efficacy than drug B

The pharmacodynamic properties of two beta1-adrenergic antagonists, Drug A and Drug B are being investigated. The two drugs are administered intravenously IV experimental animals and the effect on heart rate is recorded. It is observed that 80 mg of Drug A evoke the same decrease in heart rate as 10 mg of Drug B. Which of the following best describes the properties of Drug A?

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- C. Drug A has a lower potency than drug B
- D. Drug A has a shorter half-life than drug B
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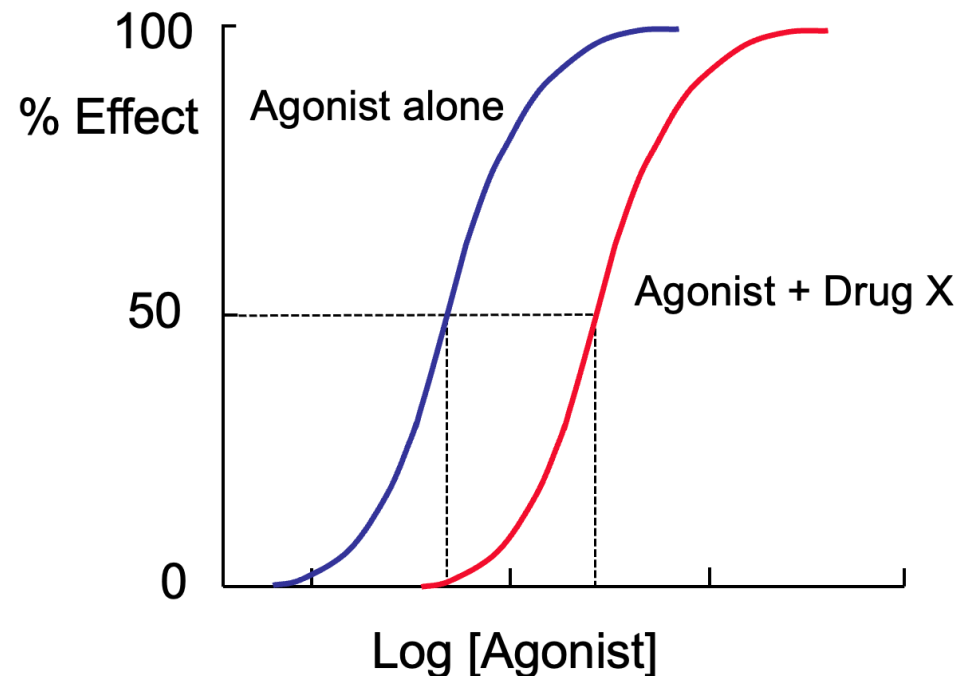
A 46-year-old man comes to the physician for a routine examination. His blood pressure is 150/94 mm Hg. Physical examination shows no abnormalities. He is diagnosed with hypertension and started on an antihypertensive drug, Drug X. The graded dose-response curve for Drug X and an agonist of Drug X is shown in the graph below. The drug on its own has no effect. Which of the following best describes the pharmacodynamic properties of this drug?

- A. Non-competitive Antagon
- B. Partial Agonist
- C. Reversible Competitive antagonist
- D. Full Agonist
- E. Physiological



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A novel drug is being studied in clinical trials to evaluate its pharmacokinetic properties. The drug is administered both orally and intravenously (IV) to healthy volunteers. The plasma concentrations of the drug at various time points are plotted on a time graph. The pharmacokinetic parameters recorded are shown in the table below. Which of the following is most likely the bioavailability of the drug?

Oral Dose (mg)	IV Dose (mg)	AUC oral	AUC IV
250 mg	250 mg	$40 \mu gh/ml$	$50 \mu gh/ml$

- A. 20%
- B. 30%
- C. 40%
- D. 60%
- E. 80%

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THE END